

Rational Functions and Asymptotes

ANSWER KEY



Finding asymptotes and intercepts

- 1 For $f(x) = \frac{x+2}{x-3}$, find:
- a the vertical asymptote $x = 3$
 - b the horizontal asymptote $y = 1$
 - c the x -intercept $(-2, 0)$
 - d the y -intercept $(0, -\frac{2}{3})$
- 2 Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.
Vertical: $x = 2$ and $x = -2$. Horizontal: equal degrees, so $y = \frac{2}{1} = 2$.
- 3 Consider $f(x) = \frac{x^2-9}{x-3}$.
- a Simplify f and identify the feature at $x = 3$. $f(x) = x + 3$ for $x \neq 3$: a hole at $(3, 6)$.
 - b Does the graph have a vertical asymptote? No — the discontinuity is removable, not an asymptote.
- 4 Find the slant (oblique) asymptote of $f(x) = \frac{x^2+1}{x-1}$.
Dividing: $f(x) = x + 1 + \frac{2}{x-1}$, so the slant asymptote is $y = x + 1$.

Behaviour near asymptotes

- 5 For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:
- a $x \rightarrow 3^+$ $f(x) \rightarrow +\infty$
 - b $x \rightarrow 3^-$ $f(x) \rightarrow -\infty$
 - c $x \rightarrow \infty$ $f(x) \rightarrow 1$
- 6 Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.
One example: $f(x) = \frac{1}{(x-1)(x+4)}$ (any numerator of degree less than 2 works).